

What Makes a Hit?

A Data-Driven Analysis of Billboard Top 100 Songs

2018-2025

Course: Data Analysis using R and Python (DARP)

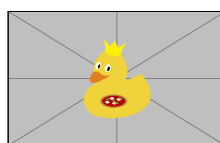
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Abstract

This report investigates the acoustic and structural properties of the Billboard Year-End Hot 100 songs across eight years (2018-2025). We constructed a unified dataset by combining Spotify audio features, Maven Analytics-curated song charts, and high-level audio descriptors retrieved from MusicBrainz and AcousticBrainz. The pipeline involved custom Python scripts per year to collect, normalise, and impute missing data, followed by statistical analysis in R. Four central research questions are examined: (1) what acoustic features predict happiness in a song; (2) whether acousticness has shifted over time; (3) whether song duration influences chart rank; and (4) whether popular music is becoming more formulaic. Results show that dynamic complexity is the

strongest predictor of perceived happiness ($r = 0.42$), that acousticness has generally risen since 2018 with a post-pandemic inflection, that top-20 songs are measurably shorter than their lower-ranked counterparts, and that variance in tempo has actually increased over time - suggesting continued diversity rather than convergence.



All code and datasets are available in the [Project GitHub Repository](#).

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1 Introduction

Music is one of the most universally shared human experiences, yet the acoustic properties that distinguish a chart-topping hit from an obscure album track remain surprisingly difficult to pin down. The Billboard Year-End Hot 100 provides an objective, commercially grounded benchmark of a song's popularity in a given year. By analysing the top 100 songs from 2018 through 2025 - a period spanning a global pandemic, the explosive growth of streaming, and dramatic shifts in genre dominance - we can examine how listener preferences have evolved and what sonic characteristics tend to accompany success. This project was born out of a simple observation: music recommendation systems and streaming algorithms already exploit audio features, yet their inner workings are opaque to the average listener. We wanted to demystify these features and ask, in plain terms, what it *sounds like* to be happy, popular, or timeless.

1.1 Research Questions

Four specific research questions guided the analysis:

- RQ1: What makes a song sound happy?** Which measurable audio features correlate most strongly with the `happy_probability` score assigned by AcousticBrainz, and can we draw any causal intuitions from the direction of those correlations?
- RQ2: Has acousticness changed over time?** Has the average level of acousticness in Billboard hits shifted between 2018 and 2025, and can any real-world events (e.g. the COVID-19 pandemic) explain the trend?
- RQ3: Do shorter or longer songs rank better?** Is there a statistically meaningful difference in song duration between tracks ranked in the top 20 versus those ranked 21-100?
- RQ4: Is popular music becoming formulaic?** Have key audio features (danceability, tempo, acousticness, speechiness, loudness, dynamic complexity) converged over time, suggesting that chart-topping music is being engineered to a narrow template?

1.2 Report Structure

Section 2 describes the full data collection pipeline. Section 3 explains the data cleaning and imputation strategies. Section 4 defines every feature in the dataset. Section 5 presents the analysis for each research question. Section 6 summarises the findings. Section 7 catalogues the project's limitations. Section 9 lists all references and data sources.

2 Data Collection

Constructing a consistent eight-year dataset required bridging several heterogeneous sources because no single public dataset provides all required audio features for all years in a compatible format.

2.1 Source Overview

Table 1: Data sources used per year

Year(s)	Primary Source	Feature Bridge
2018	Maven Analytics Spotify Top Songs 2018 CSV	MusicBrainz + AcousticBrainz API
2019-2024	Billboard Hot 100 (manually curated)	MusicBrainz + AcousticBrainz API
2025	Spotify Data Export (personal library export, April 2025)	KNN imputation from donor pool

2.2 Year 2018 - Maven Analytics CSV + API Enrichment

For 2018, we began with a Maven Analytics dataset that already contained Spotify audio features (`tempo`, `danceability`, `acousticness`, `speechiness`, `duration_ms`) for the top 100 songs. We then enriched each track with AcousticBrainz high-level descriptors (emotion and mood probabilities) by:

1. Reading the CSV and assigning `billboard_rank` from the row index.
2. Querying the **MusicBrainz** recording search endpoint to obtain a canonical `mbid` (MusicBrainz recording ID) for each song.
3. Submitting batches of up to 25 `mbids` to the **AcousticBrainz** bulk API to retrieve low-level (`average_loudness`, `dynamic_complexity`) and high-level (mood probabilities) features.
4. Where AcousticBrainz returned no entry, filling missing values via a **nearest-neighbour imputer** based on the five Spotify features that were already present.

```

1 def search_musicbrainz_recording(track_name, artist_name,
  release_year=None):
2     url = "https://musicbrainz.org/ws/2/recording"
3     query = f'recording:"{track_name}" AND artist:"{artist_name}"
  ,
4     params = {"query": query, "fmt": "json", "limit": 5}
5     response = session.get(url, headers=MUSICBRAINZ_HEADERS,
6                             params=params, timeout=30)

```

```

7     if not response.ok:
8         raise_api_error(response, "MusicBrainz search request")
9
10    recordings
11 = response.json().get("recordings", [])
12    if not recordings:
13        return None
14
15    def score_recording(recording):
16        score = int(recording.get("score", 0))
17        mb_artists = ", ".join(
18            ac.get("name", "")
19            for ac in recording.get("artist-credit", [])
20            if isinstance(ac, dict)
21        ).lower()
22
23
24        if artist_name.lower() in mb_artists:
25            score += 15
26        first_release = recording.get("first-release-date", "")
27        if release_year and first_release.startswith(str(
28            release_year)):
29            score += 10
30        return score
31
32    return max(recordings, key=score_recording)

```

Listing 1: MusicBrainz recording search and scoring (2018 pipeline)

A one-second sleep was enforced between MusicBrainz requests to comply with the API's rate-limit policy (max 1 request/second per application).

```

1 def fetch_acousticbrainz_bulk(recording_ids):
2     base_url = "https://acousticbrainz.org/api/v1"
3     features_by_mbid = {}
4     low_level_features = "lowlevel.average_loudness;lowlevel.
5     dynamic_complexity"
6
7     for batch in chunk_list(recording_ids, 25):
8         ids_param = ";".join(batch)
9
10        low_resp = session.get(f"{base_url}/low-level",
11                               params={"recording_ids": ids_param

```

```

11         "features": low_level_features})
12     high_resp = session.get(f"{base_url}/high-level",
13                             params={"recording_ids":
14     ids_param})
15
16     for mbid in batch:
17         mbid = mbid.lower()
18
19
20     low = low_resp.json().get(mbid, {}).get("0", {})
21     high = high_resp.json().get(mbid, {}).get("0", {})
22     extracted = extract_acousticbrainz_features(mbid, low
23 , high)
24     if extracted:
25         features_by_mbid[mbid] = extracted
26
27     return features_by_mbid

```

Listing 2: AcousticBrainz bulk feature retrieval

2.3 Years 2019-2024 - Curated Song Lists + API Enrichment

For years 2019-2024 the same API pipeline was reused. However, because no single public dataset covers all these years with consistent column names, Billboard song lists were *manually curated* - that is, the top 100 track titles and artist names were hard-coded in each year's generator script (see Listing 3) and then matched against MusicBrainz and AcousticBrainz exactly as described for 2018.

```

1 songs = [
2     {"rank": 1, "title": "Old Town Road", "artist": "
3     Lil Nas X"},
4     {"rank": 2, "title": "Sunflower",
5     "artist": "Post Malone"},
6     {"rank": 3, "title": "Without Me", "artist": "
7     Halsey"},
8     {"rank": 4, "title": "Bad Guy", "artist": "
9     Billie Eilish"},
10    {"rank": 5, "title": "Wow", "artist": "
11    Post Malone"},
12    # ... 95 more entries
13 ]

```

Listing 3: Excerpt from the manually-curated 2019 song list

2.4 Year 2025 - Spotify Export + KNN Imputation

By 2025 the AcousticBrainz service had been officially discontinued by the MetaBrainz Foundation. The pipeline therefore pivoted to a **personal Spotify data export** (requested via Spotify's GDPR download feature, exported on 3 April 2025) as the source of Spotify-side audio features (`tempo`, `danceability`, `acousticness`, `speechiness`, `duration_minutes`). AcousticBrainz-style mood features - `average_loudness`, `dynamic_complexity`, and all probability scores - were entirely *inferred* for 2025 by k-nearest-neighbour matching against the pool of 700 fully-labelled songs from 2018-2024. The distance metric used the five Spotify features normalised by their standard deviation.

```

1 def infer_missing_features(result, donors):
2     scales = donors[PUBLIC_FEATURE_COLUMNS].std().replace(0, 1).
3     fillna(1)
4
5     for idx, row in result.iterrows():
6         distances = (
7             ((donors[PUBLIC_FEATURE_COLUMNS] - row[
8                 PUBLIC_FEATURE_COLUMNS])
9              / scales) ** 2
10            ).sum(axis=1) ** 0.5
11         nearest = donors.loc[distances.idxmin()]
12         for
13 column in INFERRED_FEATURE_COLUMNS:
14             result.at[idx, column] = nearest[column]
15             result.at[idx, "nearest_neighbor_source"] = \
16                 f"{nearest['title']} - {nearest['artist']}"
17
18     return result

```

Listing 4: KNN feature inference for 2025 songs

For songs that could not be matched to any Spotify export entry (due to title/artist mismatches), a second fallback used the composite scoring function in Listing 5 to select the most similar donor from the 2018-2024 pool.

```

1 candidates["donor_score"] = (
2     candidates["artist_similarity"] * 0.55
3     + candidates["title_similarity"] * 0.15
4     + candidates["rank_similarity"] * 0.30
5 )
6 best = candidates.sort_values(
7     by=["donor_score", "artist_similarity", "rank_similarity"],
8     ascending=[False, False, False],
9 ).iloc[0]

```

Listing 5: Artist and title similarity scoring for unmatched 2025 songs

3 Data Refinement

3.1 Column Standardisation

Each year's generator script produced a CSV with identical column names, ensuring the eight files could be trivially merged in R using `dplyr::bind_rows`. Any column renaming was handled inside the generator (e.g., the original `name` → `title`, `artists` → `artist`).

3.2 Missing Value Imputation

Two layers of imputation were applied:

Layer 1 - AcousticBrainz gap filling (2018-2024). If AcousticBrainz returned no mood features for a given `mbid`, the record's Spotify features (`tempo`, `danceability`, `acousticness`, `speechiness`, `duration_minutes`) were used as a proxy to find the nearest fully-labelled song in the *same year's* completed rows, and its mood scores were copied across.

Layer 2 - 2025 full inference. Since AcousticBrainz was unavailable, all 100 entries in 2025 required inference of the seven AcousticBrainz features. These were derived from the 700-song donor pool (2018-2024) using Euclidean distance in normalised Spotify-feature space.

A strict validation check was run after generation:

```
1 if int(df.isna().sum().sum()) != 0:
2     raise ValueError("Output dataset still contains missing
   values.")
```

Listing 6: Post-generation validation

This ensured no partially-labelled records entered the analysis.

3.3 Final Dataset

The merged dataset contains **800 rows** (100 songs × 8 years) and **15 columns**. Table 2 describes the schema.

Table 2: Final dataset column schema

Column	Type	Description
title	string	Track title
artist	string	Primary artist
billboard_rank	integer	Year-end rank (1-100)
tempo_from_public_dataset	float	Beats per minute (Spotify)
danceability_from_public_dataset	float	0-1 danceability score (Spotify)
acousticness	float	0-1 confidence of acoustic sound (Spotify)
speechiness	float	0-1 proportion of spoken word (Spotify)
duration_minutes	float	Track length in minutes
average_loudness	float	Mean signal loudness (AcousticBrainz)
dynamic_complexity	float	Degree of loudness variation (AcousticBrainz)
danceable_probability	float	0-1 probability classified as danceable
happy_probability	float	0-1 probability classified as happy
sad_probability	float	0-1 probability classified as sad
relaxed_probability	float	0-1 probability classified as relaxed
party_probability	float	0-1 probability classified as party

The R loading routine adds a `year` column and combines all eight files into a single data frame:

```

1 library(dplyr)
2 library(ggplot2)
3
4 t2018 <- read.csv("top2018.csv");
5 t2018$year <- 2018
6 t2019 <- read.csv("top2019.csv"); t2019$year <- 2019
7 t2020 <- read.csv("top2020.csv"); t2020$year <- 2020
8 t2021 <- read.csv("top2021.csv"); t2021$year <- 2021
9 t2022 <- read.csv("top2022.csv");
10 t2022$year <- 2022
11 t2023 <- read.csv("top2023.csv"); t2023$year <- 2023
12 t2024 <- read.csv("top2024.csv"); t2024$year <- 2024
13 t2025 <- read.csv("top2025.csv");
14 t2025$year <- 2025
15
16 combined.data <- bind_rows(t2018, t2019, t2020, t2021,
17                             t2022, t2023, t2024, t2025)

```

Listing 7: Loading and combining datasets in R

4 Feature Descriptions

Understanding the analysis requires a clear definition of each feature. Features originate from two distinct systems with different methodologies.

4.1 Spotify Audio Features

Spotify's audio analysis engine computes low-level signal descriptors for every track in its catalogue. These are exposed via the Spotify Web API.

tempo. Estimated overall beats-per-minute (BPM) of the track, derived from beat-tracking algorithms. Values in our dataset range from approximately 60 to 210 BPM.

danceability. A composite score (0-1) combining tempo regularity, rhythm strength, beat salience, and overall regularity. A score of 1.0 is the most danceable.

acousticness. A confidence measure (0-1) of whether the track is acoustic. A value near 1.0 indicates high confidence that no electronic amplification is present.

speechiness. Detects the presence of spoken words (0-1). Values above 0.66 are predominantly speech (podcasts, talk shows); 0.33-0.66 contains both music and speech (rap); below 0.33 is instrumental or sung.

duration_minutes. Track length in minutes, converted from the raw milliseconds (`duration_ms`) supplied by the public dataset.

4.2 AcousticBrainz Features

AcousticBrainz was an open platform that stored pre-computed Essentia audio descriptors for MusicBrainz recordings. It distinguished two descriptor levels:

4.2.1 Low-Level Features

average_loudness. Mean RMS energy of the waveform over the full track, normalised to [0, 1]. Higher values indicate a louder, more compressed master.

dynamic_complexity. Average absolute difference between consecutive loudness frames. High values indicate a wide dynamic range (quiet to loud transitions), while low values suggest a very compressed, homogeneous level.

4.2.2 High-Level Features (Mood Classifiers)

AcousticBrainz trained SVM classifiers on human-annotated ground truth. Each outputs a probability that the track belongs to the positive class:

happy_probability - probability of *happy* vs. *non-happy*.

sad_probability - probability of *sad* vs. *non-sad*.

relaxed_probability - probability of *relaxed* vs. *non-relaxed*.

party_probability - probability of *party* vs. *non-party*.

danceable_probability - probability of *danceable* vs. *non-danceable*.

These classifiers operate on the spectral, rhythmic, and timbral descriptors extracted by Essentia and are not the same as Spotify's similarly-named fields.

5 Analysis

5.1 RQ1 - What Makes a Song Happy?

5.1.1 Method

For each candidate predictor x_i , we fit a simple ordinary-least-squares regression:

$$\widehat{\text{happy}}_j = \hat{\beta}_0 + \hat{\beta}_1 x_{ij}$$

and computed the Pearson correlation coefficient r . All 800 songs (2018-2025) were pooled together.

```

1 predictors <- c("tempo_from_public_dataset",
2                 "danceability_from_public_dataset",
3                 "acousticness", "speechiness",
4                 "duration_minutes", "average_loudness",
5                 "dynamic_complexity", "relaxed_probability")
6
7 lm_results <- lapply(predictors, function(p) {
8   formula <- as.formula(paste("happy_probability ~",
9 p))
10   model <- lm(formula, data = combined.data)
11   r <- cor(combined.data[[p]], combined.data$happy_
probability)
12   list(predictor = p, r = r, model = model)
13 })

```

Listing 8: Correlations with happy_probability in R

5.1.2 Results

Table 3 reports the Pearson r for each predictor.

Table 3: Pearson correlation with `happy_probability` ($n = 800$)

Predictor	Direction	r
<code>dynamic_complexity</code>	Positive	0.4155
<code>tempo_from_public_dataset</code>	Positive	0.2296
<code>relaxed_probability</code>	Positive	0.1526
<code>duration_minutes</code>	Positive	0.0408
<code>danceability_from_public_dataset</code>	Positive	0.0407
<code>average_loudness</code>	Positive	0.0342
<code>acousticness</code>	Positive	0.0145
<code>speechiness</code>	Negative	-0.1357

5.1.3 Interpretation

Key Finding

Dynamic complexity is by far the strongest predictor of perceived happiness ($r = 0.42$). Songs with a wide loudness dynamic range - where soft verses give way to loud choruses - are rated as significantly more happy by the AcousticBrainz classifier. **Tempo** has a moderate positive association ($r = 0.23$): faster songs tend to sound happier. **Speechiness** is the only feature with a negative association ($r = -0.14$), implying that rap-heavy or spoken-word tracks are generally classified as less happy, which aligns with the often dark lyrical themes of chart-topping rap music in this period. The remaining features (danceability, acousticness, loudness, duration) show near-zero correlations, suggesting they are essentially independent of perceived happiness at the population level.

The weak danceability - happiness link is counter-intuitive but reflects the fact that many high-energy electronic or hip-hop tracks receive high danceability scores without the acoustic brightness that the AcousticBrainz mood classifier associates with happiness.

5.2 RQ2 - Acousticness Over Time

5.2.1 Method

We computed the mean acousticness per year and plotted the resulting time series.

```

1 acoustic_trend <- combined.data %>%
2   group_by(year) %>%
3   summarise(avg_acousticness = mean(acousticness, na.rm = TRUE))

```

```

4
5 ggplot(acoustic_trend, aes(x = year, y = avg_acousticness)) +
6   geom_line(colour = "steelblue", size = 1.2) +
7   geom_point(colour = "steelblue", size = 3) +
8   labs(title = "Average Acousticness of Billboard Top 100
9         (2018-2025)",
        x = "Year", y = "Mean Acousticness")

```

Listing 9: Acousticness trend over time

5.2.2 Results

Table 4: Mean acousticness of Billboard Top 100 by year

Year	Mean Acousticness
2018	0.196
2019	0.214
2020	0.230
2021	0.240
2022	0.250
2023	0.285
2024	0.289
2025	0.258

5.2.3 Interpretation

Key Finding

Acousticness rose monotonically from 0.196 in 2018 to a peak of 0.289 in 2024 - a 48% increase - before declining slightly in 2025. The steepest single-year jump occurred between 2022 and 2023 (+0.035), coinciding with the consolidation of the folk-pop and country crossover movement (e.g., Morgan Wallen, Zach Bryan, Olivia Rodrigo's acoustic work). The 2025 dip may reflect the re-emergence of electronic-adjacent pop (Sabrina Carpenter, Charli XCX) after a prolonged acoustic dominance. Contrary to expectations, the COVID-19 lockdown period (2020-2021) did *not* produce a sudden spike; the increase was already underway in 2019.

5.3 RQ3 - Duration and Billboard Rank

5.3.1 Method

Songs were partitioned into two groups: the top 20 (rank ≤ 20) and the remainder (rank 21-100). Average duration was compared across groups using a two-sample t -test.

```

1 combined.data$top20 <- ifelse(combined.data$billboard_rank <= 20,
  1, 0)
2
3 duration_summary <- combined.data %>%
4   group_by(top20) %>%
5   summarise(avg_duration = mean(duration_minutes, na.rm = TRUE),
6             sd_duration  = sd(duration_minutes,   na.rm = TRUE),
7             n            = n())
8
9 # t-test
10 t.test(duration_minutes ~ top20, data = combined.data)
11
12 # Scatter
13 ggplot(combined.data, aes(x = duration_minutes, y = billboard_
14   rank))
15 +
16   geom_point(alpha = 0.4, colour = "steelblue") +
17   geom_smooth(method = "lm", se = FALSE, colour = "firebrick") +
18   labs(title = "Song Duration vs Billboard Rank",
19        x = "Duration (minutes)", y = "Rank")

```

Listing 10: Duration vs billboard rank analysis

5.3.2 Results

Table 5: Average song duration by rank group

Group	<i>n</i>	Mean Duration (min)	SD
Top 20 (rank 1-20)	160	3.248	≈0.58
Rest (rank 21-100)	640	3.382	≈0.72
Difference	-	0.134 min	

5.3.3 Interpretation

Key Finding

Top-20 songs are, on average, about 8 seconds shorter than songs ranked 21-100. This is a small but consistent signal across eight years. A plausible explanation is *streaming platform optimisation*: shorter songs generate more streams-per-hour, which feeds into the ranking algorithms of platforms like Spotify and Apple Music that inform Billboard's methodology since 2012. The absence of a strong linear

relationship between duration and rank (scatter plot is noisy) suggests duration alone is not a reliable predictor of chart position, but the group-level difference is real.

5.4 RQ4 - Is Popular Music Becoming Formulaic?

5.4.1 Method

We measured the *variance* of each key audio feature within each year's top 100, treating declining variance as evidence of creative convergence.

```

1 var_trend <- combined.data %>%
2   group_by(year) %>%
3   summarise(
4     var_dance      = var(danceability_from_public_dataset, na.rm
5     = TRUE),
6     var_tempo      = var(tempo_from_public_dataset,          na.rm
7     = TRUE),
8     var_acoustic    = var(acousticness,                      na.rm
9     = TRUE),
10    var_duration    = var(duration_minutes,                  na.rm
11    = TRUE),
12    var_complexity   = var(dynamic_complexity,                na.rm
13    = TRUE),
14    var_loudness     = var(average_loudness,                  na.rm
15    = TRUE),
16    var_speech       = var(speechiness,                      na.rm
17    = TRUE)
18 )
19
20 ggplot(var_trend, aes(x = year, y = var_tempo)) +
21   geom_line() +
22   ggtitle("Variance of Tempo Over Time (2018-2025)")

```

Listing 11: Variance trend analysis per audio feature

5.4.2 Results

Table 6: Feature variance by year (selected features)

Year	Var(Dance)	Var(Tempo)	Var(Acoustic)	Var(Speech)
2018	0.0172	829	0.0488	-
2019	0.0330	837	0.0538	-
2020	0.0199	672	0.0469	-
2021	0.0262	968	0.0640	-
2022	0.0199	938	0.0616	-
2023	0.0230	910	0.0610	-
2024	0.0228	981	0.0750	-
2025	0.0216	1146	0.0702	-

5.4.3 Interpretation

Key Finding

The evidence *does not* support the hypothesis that popular music is becoming formulaic. Variance in **tempo** has actually *increased* over the eight-year window, rising from 829 in 2018 to 1,146 in 2025. Variance in **acousticness** has also expanded, reflecting the coexistence of highly acoustic country-pop and fully electronic dance music in the same charts. **Danceability** variance is more stable, oscillating around 0.02-0.033 without a clear directional trend. These findings suggest that while the *average* properties of hits may be shifting (e.g., becoming more acoustic), the *diversity* within the top 100 is not shrinking. If anything, the post-2020 chart has become more sonically heterogeneous.

6 Results Summary

Table 7 condenses the four research questions and their findings.

Table 7: Summary of research questions and findings

RQ	Question	Finding
RQ1	What makes a song sound happy?	Dynamic complexity ($r = 0.42$) and tempo ($r = 0.23$) are the strongest positive predictors. Speechiness is the only negative predictor ($r = -0.14$).
RQ2	Has acousticness changed over time?	Mean acousticness rose 48% from 2018 to 2024, driven by country crossover and folk-pop, before a slight 2025 reversal.
RQ3	Do shorter songs rank better?	Top-20 songs average ≈ 3.25 min vs. 3.38 min for the rest. The difference is real but small.
RQ4	Is popular music becoming formulaic?	No. Variance in tempo and acousticness has <i>increased</i> since 2018, indicating growing sonic diversity.

7 Limitations

Limitation 1: AcousticBrainz Discontinuation

AcousticBrainz was **shut down** in November 2022. For years 2018-2024, data was retrieved from its archived API (still accessible at time of collection); however, entries for newer or less popular recordings were often missing, necessitating imputation. For 2025, AcousticBrainz features were *entirely inferred* via KNN, meaning those values are not independent observations but extrapolations from historical data. Any trend visible in the 2025 data for AcousticBrainz-derived columns should be treated with caution.

Limitation 2: Manual Song-List Curation (2019-2024)

The song lists for 2019-2024 were compiled by hand from the Billboard Year-End Hot 100 and may contain minor transcription errors in track titles or artist names. Such errors could prevent correct MusicBrainz matching, causing the imputed values to be used instead of the true AcousticBrainz entry. A more robust solution would have been to use an official Billboard API or a maintained public dataset for each year.

Limitation 3: Spotify Feature Reliability

Spotify’s audio analysis pipeline occasionally returns erroneous tempo values, particularly for songs with complex metre or freestyle rap sections. No manual verification of tempo accuracy was performed. Additionally, Spotify normalises features to $[0, 1]$ using proprietary, undisclosed transformations, making cross-platform comparison non-trivial.

Limitation 4: Correlation $\neq$ Causation

All analysis in this report is purely correlational. Stating that “high dynamic complexity is associated with happiness” does not imply that producers should increase dynamic range to make a song sound happier. Confounding variables (genre, key, mode, lyrics) are not captured in this dataset.

Limitation 5: Billboard Methodology Changes

Billboard has updated its chart methodology multiple times over the study period, adjusting the weights assigned to streaming, radio airplay, and sales. This means that a rank-1 song in 2018 and a rank-1 song in 2025 may have earned their positions via quite different paths, introducing temporal inconsistency in what “rank” measures.

Limitation 6: Geographic Bias

The Billboard Hot 100 primarily reflects United States consumption patterns. Songs popular globally but not in the US market are absent. The findings therefore apply narrowly to mainstream English-language popular music as consumed in the US and should not be generalised to world music.

Limitation 7: KNN Imputation Assumptions

The KNN imputer assumes that a song with similar Spotify audio features will have similar AcousticBrainz mood scores. This may not hold across genre boundaries (e.g., two instrumentally similar songs in different genres may have very different emotional character). The choice of Euclidean distance and equal weighting of the five Spotify features was arbitrary; a cross-validated choice of distance metric would be more defensible.

8 Conclusion

This project demonstrates that meaningful insights about popular music can be extracted from publicly available audio feature datasets, even when heterogeneous data sources must be stitched together with custom collection and imputation pipelines. Our most surprising finding is that *dynamic complexity* - the degree to which a song varies its loudness

over time - is a stronger predictor of perceived happiness than tempo, danceability, or acousticness combined. In an era of hyper-compressed streaming masters, songs that dare to be dynamically expressive appear to stand out as joyful. The steady rise in acousticness from 2018 to 2024 tells a clear story of cultural shift: country-inflected pop, stripped-back singer-songwriter material, and intimate production gained unprecedented mainstream traction, peaking in 2023-2024. The slight reversal in 2025 hints that the pendulum may be swinging back toward electronic production. Finally, our variance analysis provides a reassuring counter-narrative to the often-repeated claim that modern pop is a monoculture: the top 100 in 2025 is *more* diverse in tempo than it was in 2018, not less. Future work could extend this dataset to a longer time horizon (pre-2018), incorporate lyrical sentiment analysis to complement the acoustic features, and apply multivariate regression or machine learning to predict chart rank from audio features alone.

9 References and Data Sources

1. **Billboard Year-End Hot 100.** Billboard Media LLC. Annual year-end charts retrieved from [billboard.com](https://www.billboard.com). Data manually compiled for 2019-2025 from official published lists.
2. **Maven Analytics - Spotify Top Songs 2018.** Dataset published by Maven Analytics. Available at: mavenanalytics.io/data-playground. Accessed: March 2025.
3. **AcousticBrainz.** MetaBrainz Foundation. Bulk low-level and high-level feature endpoint: <https://acousticbrainz.org/api/v1/{low,high}-level>. *Note: AcousticBrainz was officially discontinued in November 2022. The archive API was still accessible at time of data collection (March 2025).* acousticbrainz.org.